Data Mining Syllabus

Data Mining and Analytics (SRM Institute of Science and Technology)

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Course Code	21CSE355T	Course Name	DATA MINING AND ANALYTICS	Course Category	E	PROFESSIONAL ELECTIVE	L	T	P	C
							2	1	0	3

Pre-requisite Courses	*Nil*	Co- requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	*School of Computing*	Data Book / Codes / Standards	*Nil*		

Course Learning Rationale (CLR):	*The purpose of learning this course is to:*
CLR-1:	*introduce the basic concepts of pattern discovery and data preparation*
CLR-2:	*understand the importance of Association and Correlation Algorithms*
CLR-3:	*comprehend and apply various Classifiers*
CLR-4:	*work with the foundation for Clustering*
CLR-5:	*perform Outlier Analysis and Explore a data mining tool*

Course Outcomes (CO):	*At the end of this course, learners will be able to:*	Program Outcomes (PO)												Program Specific Outcomes		
		1	2	3	4	5	6	7	8	9	10	11	12			
		Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO-1	PSO-2	PSO-3
CO-1:	*do the preprocessing of data before mining of data for patterns*	*1*	*2*	-	-	-	-	-	-	-	-	-	-	*2*	-	-
CO-2:	*make use of Association and Correlations Algorithms for framing association rules*	*1*	*2*	-	-	*3*	-	-	-	-	-	-	-	*2*	-	-
CO-3:	*apply as well as Compare the performance of various classifiers*	*1*	*2*	-	-	*3*	-	-	-	-	-	-	-	*2*	-	-
CO-4:	*utilize different Clustering algorithms for generalization*	*1*	-	-	-	*3*	-	-	-	-	-	-	-	*2*	-	-
CO-5:	*identify Outliers in the data given*	*1*	*2*	-	-	*3*	-	-	-	-	-	-	-	*2*	-	-

Unit-1 - Data Mining Introduction	***9 Hour***
Introduction: Kinds of Data- Kinds of Patterns-Data Objects and Attribute Type- Data Visualization -Data Preprocessing: Data cleaning, Data Integration, Data Transformation, Data Discretization and Data Reduction: Attribute Subset Selection-Histograms, Clustering, Sampling	
Unit-2 - Associations and Correlations	***9 Hour***
Market Basket Analysis – Apriori Algorithm – Mining Frequent Itemsets without Candidate Generation – Mining Frequent Itemsets Using Vertical Data Format – Mining Closed Frequent Itemsets – Mining Multilevel Association Rules – Mining Multidimensional Association Rules – Correlation Analysis – Constraint-Based Association Mining	
Unit-3 - Classification and Prediction	***9 Hour***
Basic Concepts- Decision Tree Induction-Attribute selection Measures-ID3 and CART algorithms, Tree Pruning-Bayes Classification Methods: Bayes" Theorem, Naive Bayesian Classification - Classification by Backpropagation- Support Vector Machines-Lazy learners: KNN-Metrics for evaluating classifier performance-Techniques to improve classification accuracy-Prediction: Regression Analysis	
Unit-4 - Cluster Analysis	***9 Hour***
Cluster Analysis: Partitioning Methods- Hierarchical Methods: Agglomerative versus Divisive Hierarchical Clustering-Probabilistic Model based Clustering - BIRCH, DBSCAN, STING, CLIQUE Techniques- Evaluation of clustering Techniques	
Unit-5 - Outliers and Statistical Approaches in Data Mining	***9 Hour***
Introduction to outliers, Challenges in detecting Outliers,Outlier Detection Methods - Supervised, Semisupervised, Unsupervised- Statistical Data Mining approaches - Data mining in Recommender Systems,Data mining for Intrusion Detection, Data Mining for Financial Analysis	

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Learning Resources	1. *Jiawei Han and Micheline Kamber, "Data Mining Concepts and Techniques", Third Edition, Elsevier, 2012* 2. *Ian H. Witten, Eibe Frank and Mark A. Hall "Data Mining: Practical Machine Learning Tools and Techniques", Fourth Edition, Elsevier, 2017.*

Learning Assessment

	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (50%)		Life-Long Learning CLA-2 (10%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	15%	-	15%	-	15%	-
Level 2	Understand	25%	-	20%	-	25%	-
Level 3	Apply	30%	-	25%	-	30%	-
Level 4	Analyze	30%	-	25%	-	30%	-
Level 5	Evaluate	-	-	10%	-	-	-
Level 6	Create	-	-	5%	-	-	-
	Total	100 %		100 %		100 %	

Course Designers

Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
Expert Member from TCS		
1. Dr. T. Ruso,Senior Project Lead, HCL Technologies, Chennai	*1. Khanna Nehemiah, Associate Professor, Anna University Chennai*	

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